Arpita Waghmare

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Profile

Computer Science undergraduate focused on **Artificial Intelligence, Machine Learning, Data Science, and Backend Development**. Experienced in building ML systems, AI-agent workflows, REST APIs, and document intelligence applications using **Python, JavaScript, Node.js, LangGraph, Langflow, and Explainable AI**.

Education

Symbiosis Institute of Technology, Nagpur — **Nagpur, Maharashtra**
B.Tech in Computer Science Engineering; CGPA: 8.2 — *Third Year – 5th Semester*

Indian Institute of Technology Madras — **Online**
B.S. in Data Science and Applications — *Ongoing*

Technical Skills

Programming: Python, Java, C, JavaScript

AI/ML: Machine Learning, Explainable AI, Scikit-learn, XGBoost, Random Forest, ANN, Fuzzy KNN, SHAP

AI Agents & LLM: LangGraph, Langflow, LangChain, Groq, Prompt Engineering

Backend & Web: Node.js, REST APIs, HTML, CSS, JavaScript, Gradio

Databases & Tools: SQL, DBMS, Redis, RabbitMQ, Celery, Git, GitHub, VS Code, Postman, Google Colab, Hugging Face

Projects

DocuMind – Intelligent Document Organization Assistant GitHub

- Developed an AI-powered document intelligence system for analyzing uploaded PDFs, organizing information, and generating context-aware quizzes.
- Integrated **LangGraph/Langflow** for AI-agent workflow orchestration and **Groq** for LLM-powered document processing.
- Built a Gradio-based interface and explored asynchronous processing using **Redis, RabbitMQ, and Celery**.

Todo List REST API – Node.js GitHub

- Built a REST API from scratch using **pure Node.js without Express**, implementing HTTP methods, routing, request handling, and in-memory data management.
- Added task priorities, due dates, automatic overdue detection, and task sorting.
- Created structured API documentation and setup instructions in the project README.

Parkinson's Disease Prediction using Voice Data GitHub

- Developed a machine learning pipeline to predict Parkinson's Disease from voice-based features using **Fuzzy K-Nearest Neighbors**.
- Applied **Explainable AI with SHAP** to interpret model predictions and identify influential features.
- Explored Digital Twin concepts for continuous patient-state monitoring and personalized analysis.

Hybrid ANN–Random Forest Climate Forecasting GitHub

- Designed a hybrid forecasting model combining **Artificial Neural Networks and Random Forest** for climate prediction.
- Applied feature engineering, temporal data splitting, model evaluation, and seasonal bias correction to improve forecasting reliability.

Certifications

Oracle Cloud Infrastructure 2025 – Certified Data Science Professional – Cloud computing, data processing, machine learning workflows, and model evaluation using Oracle Cloud.

Google – Gemini for End-to-End SDLC (2025) – Software development lifecycle, requirement analysis, AI-assisted development, and deployment workflows.

Position of Responsibility

Core Member – IEEE Student Chapter – Contributed to technical workshops, peer-learning initiatives, collaborative activities, and technology-focused events.